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WASHINGTON, DC

Is it just allergic rhinitis? Predictive modeling to identify biomarkers associated with asthma comorbidity using both clinical and airway microbiome data

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Abstract

Our study developed machine learning models to predict whether patients with allergic rhinitis (AR) also had asthma comorbidity (ARAS), given patients' clinical measures and airway microbiome abundances. Logistic regression, stepwise selection, LASSO, and random forest yielded models with between 70-78% accuracy; the highest was achieved by a random forest which used a subset of predictors from stepwise selection. Interpreting the resulting models yielded a subset of potential key predictors that included both clinical measures and microbiomes.

Introduction

- Asthma prevalence is 11.5% worldwide, and 4.13% in Portugal (Song et al., 2022); allergic rhinitis (AR) prevalence worldwide is 18.1% (Savouré et al., 2022).
- Asthma accounts for 1 in 250 deaths globally (Masoli et al., 2014)
- While asthma is a disease of the lower respiratory tract and AR is a disease of the upper respiratory tract; they have shown to be linked, and frequently coexist (Compalati et al., 2010); asthma is often more severe when coexisting with AR.
- Studies have demonstrated that airway microbiota are involved in the onset and progression of both asthma and allergic rhinitis (Pérez-Losada et al., 2023).
- Our aim was to develop predictive models to distinguish between AR and ARAS using microbiome and clinical measures. Identifying key predictors may be useful in better understanding interrelationships between AR and asthma.

Data

ASMAPORT study

- Cross-sectional study of healthy children and adults in Portugal suspected to have allergic rhinitis and/or asthma
- Data collected 2018-2020
- Our study focused on the subset of 215 participants presenting with either AR alone or ARAS (both AR and asthma)
- n = 215, with n = 130 complete cases

Outcome Variable:

asthma_rhinitis:
AR (n=32): pt has allergic rhinitis
ARAS (n=98): pt has asthma accompanied by allergic rhinitis

Data cleaning:

Removed clinical variables where non-missing data was sparse (>50% missing). Removed binary variables where data set did not clearly distinguish between reference level (e.g., 0) and missing.

Exposure Variables retained after data cleaning:

Clinical Variables (n=17):

Numeric variables:
age: age (years)
BMI: Body mass index (kg/cm^2)
pre-bronchial dilation tests (4): FVC: Forced Vital Capacity; Fev1: Forced expiratory volume; MMEF75_25: Maximal Mid-Expiratory Flow; PEF: Peak expiratory flow
post-bronchial dilation tests (4): FVC, Fev1, MMEF75_25, PEF
exhaled nitric oxide level

Categorical variables:
TCUT_pos: Positive IgE test for aeroallergens {0, 1}
season: {winter, spring, summer, fall}
gender: {male, female}

Microbiome Abundance Variables (n=21):

Normalized abundance values for 4 phyla, 17 genera, 1 unknown

Methods

Method selection considerations:

We considered the following when selecting machine learning methods:

- Binary outcome; levels were AR and ARAS
- The number of complete observations in our data set (130) was not especially large compared to the number of covariates (38). We also sought to identify a subset of key predictors from among the available variables. Thus, we favored methods which would provide subset selection.
- Favored methods with higher interpretability

Statistical analysis software

- R version 4.2.3
- R libraries: caret, glmnet, randomForest

Logistic Regression:

- Optimal threshold selected using maximum Cohen's Kappa (κ). We chose to measure accuracy where κ was maximized because κ balances specificity and sensitivity
- 10-fold cross validation to estimate test error

Stepwise Regression (both-direction):

- Both-direction stepwise regression (base::step) adds predictors to the model sequentially, then removes predictors that no longer provide improvement in fit.
- BIC (*k=log(n)*) was used as metric to select the optimal model. BIC was favored over AIC because it imposes more of a penalty on larger models.
- Accuracy estimated using custom-coded 10-fold cross-validation, maximizing Cohen's κ for each fold.

LASSO Regression:

- 10-fold cross validation to determine test accuracy
- Lambdas were computed based on glmnet defaults of nlambda=100 and lambda.min.ratio=1e-04.
- Test accuracy for minimum lambda was selected.

Random Forest with full model and Random Forest with stepwise subset:

- Variable subset selected using stepwise regression was used as the basis for Random Forest
- 500 trees were run, as OOB error rate was plotted and observed to stabilize after ~50 trees.
- Number of predictors to try at each split (mtry) was tuned over a range of 1 to 10.
- Accuracy was retrieved at the "best tune" value of mtry.
- Variable Importance Plots provided ranking of variable importances based on mean decrease in Gini index.

Results

Table 1: Variable significance, retention, and importance across machine learning models for predicting AR versus ARAS

	Model				
Variable	**Logistic (* = p<0.05)**	**Stepwise Selection**	**LASSO**	**Full Model Random Forest (top 10 variables[#])**	**Stepwise Subset Random Forest (top 10 variables[#])**
Firmicutes (phylum)	✔	✔	✔		4
Proteobacteria (phylum)	✔*	✔			8
Actinobacteria (phylum)	✔*	✔	✔		5
Bacteroitoda (phylum)	✔*	✔		3	2
Fusobacteriota (genus)	✔	✔			3
Streptococcus (genus)	✔		✔	8	
Staphylococcus (genus)	✔			4	
Dolosigranulum (genus)	✔			5	
Haemophilus (genus)	✔			9	
Moraxella (genus)	✔	✔	✔		9
Corynebacterium (genus)	✔				
Lawsonella (genus)	✔				
Veillonella (genus)	✔			2	
Neisseria (genus)	✔		✔	6	
Peptoniphilus (genus)	✔				
Finegoldia (genus)	✔		✔		
Anaerococcus (genus)	✔		✔	10	
Leptotrichia (genus)	✔		✔		
unclassified bacteria	✔		✔		
Age	✔				
Gender	✔		✔		
BMI	✔				
TCUT+	✔*		✔		
FVC.pre	✔		✔		
Fev1.pre	✔				
MMEF75_25.pre	✔			7	
PEF.pre	✔				
FVC.post	✔*				
Fev1.post	✔*	✔			7
MMEF75_25.post	✔*	✔	✔		6
PEF.post	✔				
Bronchodilation Value	✔				
Bronchodilation Positive	✔		✔		
Nitric Oxide	✔*	✔	✔	1	1
Season == spring	✔*	✔	✔		10
10CV Model Accuracy	**0.6956**	**0.7297**	**0.7615**	**0.7556**	**0.7769**

[#]top ten variables in random forest models determined by mean decrease in Gini index

Discussion

- Given the data's high dimensionality relative to the number of observations, it is not surprising that dimensionality reduction via stepwise selection and LASSO resulted in models with improved accuracy (0.7297 and 0.7615, respectively) compared to logistic regression on the full model (0.6956). Further gains were achieved by using the reduced model in a random forest (0.7769). While asthma is likely associated with a variety of interacting factors, dimensionality reduction may have out-selected more "noisy" variables, improving predictive value.
- Pérez-Losada et al. (2023) conducted Wilcoxon tests to identify phyla/genera that differed between AR and ARAS. Among phyla, *Firmicutes* showed statistically significant differences; among genera, *Anerococcus* and *Staphylococcus* showed statistically significant differences. *Firmicutes* was retained in our models after stepwise selection and after LASSO; *Anerococcus* was only retained by LASSO; neither retained *Staphylococcus*. A possible reason for the differences is that our study included clinical variables; some correlation between bacterial abundances and certain clinical variables may have caused phyla/genera to be dropped during variable selection. Use of machine learning techniques to distinguish between AR and ARAS was not found in the literature.
- *season* was originally coded with 4 levels, however our initial LASSO regression resulted in a model that only partially retained the *season* variable. We attempted to implement group LASSO using the *gglasso* package, but were not successful in having the package recognize the binary outcome. Preliminary logistic regression indicated that *season='spring'* is significant, so we consolidated this variable into two levels.

References

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Link to Video Presentation

https://youtu.be/ALEO66bmr1E